Design of an Optimal Sliding Surface for 2-DOF Twin Rotor MIMO System

Lisy E.R, M.Nandakumar,* Anasraj R.†
Electrical And Electronics Engg,
G.E.C Thrissur, Kerala, India 680009
*lisyjhn@yahoo.com
† mnkumar@gectcr.ac.in

***Abstract*—This paper proposes the application of quadratic minimization technique for the design of optimal sliding surface. The optimal sliding surface enables the system trajectory to be robust by designing a suitable Sliding Mode Controller (SMC). SMC is designed using state feedback linearization technique. The above strategy is applied for 2-degree of freedom Twin Rotor MIMO system (2- DOF TRMS).The simulation results reveal the effectiveness of the proposed design approach.**

Keywords.sliding mode control (SMC), sliding surface, 2-DOF Twin Rotor MIMO system (2- DOF TRMS). Quadratic minimization technique.

I. Introduction

The twin Rotor MIMO system (TRMS) [1] is a laboratory set up designed for control experiments. The behavior of TRMS resembles that of a helicopter whose performance is described as highly unstable nonlinear dynamics. The TRMS unit consists of two rotors placed on a beam together with a counter balance. The beam is pivoted on its base in such a way that it can rotate freely both horizontal and vertical planes. The control inputs to TRMS are supply voltages applied to its dc motors. The outputs of TRMS are angular positions of beam that is pitch and yaw angles denoted by ψ and ϕ. The performance study of TRMS is carried out using model available in [1] and [2]. The outputs (pitch angle and yaw angle) as mentioned above are to be controlled independently. This is done using decoupling approach mentioned in [1] and [3]. The decoupling can also be done using [4]. Now, the designer is faced with the design of the controller which gives optimal performance. Among the various available controllers, sliding mode controller (SMC) is best suited for the system to be robust irrespective of internal parameter changes and external disturbances. The design of SMC has two steps [5]. The first step is to define switching surface and second step is to define controller.

In this paper, highly robust switching surface is proposed. The switching surface is defined using Quadratic minimization as in [6]. The switching control is added with equivalent control obtained from feedback linearization [7] to get highly robust controller.

The remaining portion is arranged as follows. In Section II, an overview of the system parameters and decoupling are provided. In Section III surface design is given. In Section IV, sliding mode controller design has been described. In Section V the sliding mode performance is presented. Finally concluding remarks are made in Section VI. .

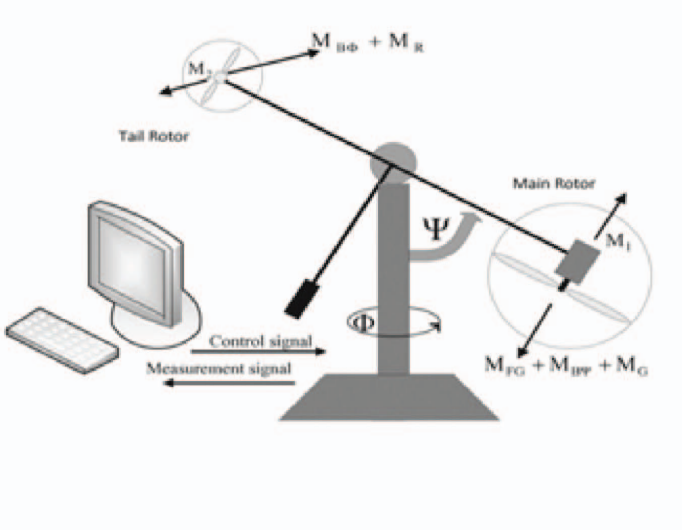


Fig. 1. TRMS system.

II. OVERVIEW OF 2-DOF TRMS

As shown in Fig. 1 , the TRMS is a mechanical unit consisting of two rotors, one main rotor and another tail rotor placed on a beam together with a counter balance [2]. The main rotor produces a lifting force allowing the beam to rise vertically while the tail rotor is used to make the beam to turn left and right around the horizontal axis. The following momentum equation can be derived for the vertical movement as it appears in [1] and [2]. The moment of inertia of main motor is given by

$$\ddot{\psi} I_1 = M_1 - M_{FG} - M_{B\psi} - M_G \tag{1}$$

where $\ddot{\psi}$ is the pitch angle acceleration, M_1 is the non-linear static characteristic of main rotor,M_{FG} is the gravity momentum, $M_{B\psi}$ is frictional force momentum and M_G is gyroscopic momentum. Also the nonlinear static characteristic

978-1-4799-7862-5/15/$31.00 © 2015 IEEE
Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on November 04,2026 at 22:37:49 UTC from IEEE Xplore. Restrictions apply.

M_1 is related with the torque τ_1 developed in main rotor as:

$$M_1 = a_1 {\tau_1}^2 + b_1 \tau_1 \tag{2}$$

where a_1 and b_1 are constants. The gravity momentum M_{FG} is related with the pitch angle ψ as:

$$M_{FG} = M_g \sin(\psi) \tag{3}$$

The frictional force momentum $M_{B\psi}$ is related to friction momentum function parameter of main rotor and tail rotor, pitch angle velocity $\dot{\psi}$ and yaw angle $\dot{\phi}$ velocity as:

$$M_{B\psi} = B_{1\psi}\dot{\psi} - \frac{0.0326}{2}\sin 2\psi\dot{\phi}^2 \tag{4}$$

The gyroscopic momentumM_G is related to yaw angle velocity, pitch angle, nonlinear static characteristic of main rotor and gyroscopic momentum parameter as:

$$M_G = K_{gy} M_1 \dot{\phi} \cos\psi \tag{5}$$

Similarly, the momentum equation for the horizontal movement is given as:

$$\ddot{\phi} I_2 = M_2 - M_{B\phi} - M_R \tag{6}$$

where $\ddot{\phi}$ is the yaw angle acceleration,M_2 is the nonlinear static characteristic of tail rotor,$M_{B\phi}$ is frictional force momentum and M_R is cross reaction momentum. The non linear static characteristic of tail rotorM_2 is related with torque τ_2 developed in the tail rotor as:

$$M_2 = a_2 {\tau_2}^2 + b_2 \tau_2 \tag{7}$$

where a_2 and b_2 are constants. Also the frictional force momentum $M_{B\phi}$ for horizontal movement is related to yaw angle velocity as:

$$M_{B\phi} = B_{2\phi}\dot{\phi} \tag{8}$$

The cross reaction momentum M_R is approximated by the following equation

$$M_R = K_c \frac{T_o s + 1}{T_p s + 1} M_1 \tag{9}$$

where T_P and T_O represent the cross reaction momentum parameters. The output and state vectors of 2 DOF TRMS are given by

$$y = \begin{Bmatrix} \psi & \phi \end{Bmatrix}^T \tag{10}$$

$$X = \begin{Bmatrix} \psi & \dot{\psi} & \phi & \dot{\phi} & \tau_1 & \tau_2 \end{Bmatrix}^T \tag{11}$$

where ψ is pitch angle and ϕ is yaw angle. From the above equations[1] to[9], it is obvious that the system is nonlinear. By linearising the system as per [2], the state space is represented in the following form given by

$$\dot{X} = EX + FU \tag{12}$$

where $U = \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix}$, u_1 is the input voltage applied to main rotor and u_2 is the input voltage applied to tail rotor. The maximum and minimum voltages for both main rotor and tail rotors are +2.5V and -2.5 V respectively. The matrix E represent the system matrix and the matrix F represents the input distribution matrix [2]. The output equation appears in [2] is represented as:

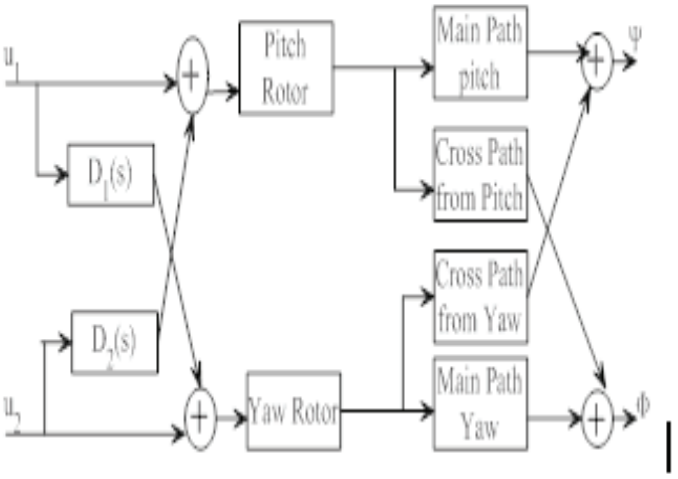


Fig. 2. TRMS with decoupler.

$$\begin{Bmatrix} \psi \\ \phi \end{Bmatrix} = \begin{Bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{Bmatrix} \begin{Bmatrix} \psi \\ \dot{\psi} \\ \phi \\ \dot{\phi} \\ \tau_1 \\ \tau_2 \end{Bmatrix} \tag{13}$$

The values of parameters in the matrices E and F are substituted as per Table. 1 of [2]. The corresponding Transfer function as given in [3] is:

$$G(S) = \begin{Bmatrix} \dfrac{1.359}{s^3 + 0.9973s^2 + 4.786s + 4.278} & 0 \\ \dfrac{1.617}{s^3 + 5.909s^2 + 4.545s} & \dfrac{3.6}{s^3 + 6s^2 + 5s} \end{Bmatrix} \tag{14}$$

From the above equation, it is obvious that there is considerable coupling between main rotors input u_1 and yaw angle ϕ. In order to eliminate this coupling a decoupler is inserted in to the TRMS. Fig. 2 represents the decoupler which is introduced to TRMS MIMO system [1]. With the introduction of decoupler, the coupling term is made to zero.Now the net transfer function of the 2 DOF TRMS system is modified as in equation (15). The modified transfer function

$$G_1(S) = \begin{Bmatrix} \dfrac{1.359}{s^3 + 0.9973s^2 + 4.786s + 4.278} & 0 \\ 0 & \dfrac{3.6}{s^3 + 6s^2 + 5s} \end{Bmatrix} \tag{15}$$

is obtained as given in [3]. (Interested readers may read the reference papers for better background).

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on November 04,2026 at 22:37:49 UTC from IEEE Xplore. Restrictions apply.

III. DESIGN OF OPTIMAL SLIDING SURFACE USING QUADRATIC MINIMIZATION

Variable structure system yields expected performance with an optimal sliding surface design. This system involves the design of switching surfaces for both pitch angle and yaw angle.The quadratic minimization approach is employed for the design of switching surfaces in both above mentioned cases.
case1. From the decoupled transfer function model (15) the transfer function of the main rotor is given by

$$\psi(s) = \frac{1.359}{s^3 + 0.9973s^2 + 4.786s + 4.278} u_1(s) \tag{16}$$

The phase variable form of the above equation is:

$$\begin{aligned} \dot{\psi_1} &= \psi_2 \\ \dot{\psi_2} &= \psi_3 \\ \dot{\psi_3} &= -4.278\psi_1 - 4.786\psi_2 - 0.9973\psi_3 + 1.359u_1 \end{aligned} \tag{17}$$

The equation (17) can be represented in matrix form as:

$$\begin{Bmatrix} \dot{\psi_1} \\ \dot{\psi_2} \\ \dot{\psi_3} \end{Bmatrix} = \begin{Bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \neg 4.278 & \neg 4.786 & \neg 0.9973 \end{Bmatrix} \begin{Bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{Bmatrix} + \begin{Bmatrix} 0 \\ 0 \\ 1.359 \end{Bmatrix} u_1 \tag{18}$$

Equation (18) represents the state model of the pitch angle. The next step is to define switching function . Switching function $S_1(t)$ is related to state vector pitch angle as given below:

$$S_1(t) = L\psi(t) \tag{19}$$

where L is switching surface matrix,$\psi(t) = [\psi_1(t)\psi_2(t)\psi_3(t)]$ The system shown in (17) is a third order system. Generally a higher order system is brought down by employing regular form approach for simplicity. For the system considered, to apply regular form a different set of state variables are employed using w(t). This is related to the conventional pitch angle state vector $\psi(t)$ as:

$$w(t) = T_r\psi(t) \tag{20}$$

where T_r is the orthogonal matrix used for coordinate transformation. Now, the regular form of equation (18) becomes

$$\begin{aligned} \dot{w_1}(t) &= A_{11}w_1(t) + A_{12}w_2(t) \\ \dot{w_2}(t) &= A_{21}w_1(t) + A_{22}w_2(t) + B_2u(t) \end{aligned} \tag{21}$$

where

$$A_{11} = \begin{Bmatrix} 0 & 1 \\ 0 & 0 \end{Bmatrix} A_{12} = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix} A_{21} = \begin{Bmatrix} \neg 4.278 & \neg 4.786 \end{Bmatrix} A_{22} = \begin{Bmatrix} \neg 0.9973 \end{Bmatrix} \tag{22}$$

Now,the associated switching function in regular form can be expressed as:

$$S_1(t) = l_1w_1(t) + l_2w_2(t) \tag{23}$$

In the switching surface, during sliding motion the switching function S_1 will be equal to zero as expressed in equation

$$l_1w_1(t) + l_2w_2(t) = 0 \tag{24}$$

,

$$w_2(t) = {l_2}^{-1}l_1w_1(t) = -M_1w_1(t) \tag{25}$$

where $M_1 = {l_2}^{-1}l_1$ The calculation of M_1 is done after the selection of quadratic performance index which is detailed in the upcoming discussion. Let the quadratic performance index be

$$J = \frac{1}{2}\int_{t_s}^{\infty} \psi^T Q\psi dt \tag{26}$$

where t_s is the time at which sliding mode commences, matrix Q is known as performance index matrix. It is selected to minimize the deviation of final state of the system from its desired value. Assuming Q as identity matrix

$$Q = \begin{Bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{Bmatrix} \tag{27}$$

The performance index Q is transformed and partitioned in compatibly with ' w(t) as

$$T_rQT_r^T = \begin{Bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22]} \end{Bmatrix} \tag{28}$$

$$Q_{21} = Q_{12}^T \tag{29}$$

where

$$Q_{11} = \begin{Bmatrix} 1 & 0 \\ 0 & 1 \end{Bmatrix} Q_{12} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} \tag{30}$$

$$Q_{21} = \begin{Bmatrix} 0 & 0 \end{Bmatrix} Q_{22} = \begin{Bmatrix} 1 \end{Bmatrix} \tag{31}$$

where 'w(t)' is a different set of state variables employed for regular form approach as per(19). Now the performance measure is represented in coordinate transform as:

$$J = \frac{1}{2}\int_{t_s}^{\infty} w_1^TQ_{11}w_1 + 2w_1^TQ_{12}w_2 + w_2^TQ_{22}w_2 \tag{32}$$

The above equation can be expressed as

$$\begin{aligned} J =& \frac{1}{2}\int_{t_s}^{\infty} w_1^T\hat{Q}w_1 + \\ & (w_2 + Q_{22}Q_{21}w_1)Q_{22}(w_2 + Q_{22}^{-1}Q_{21}w_1) \end{aligned} \tag{33}$$

where

$$\hat{Q} = Q_{11} - Q_{12}Q_{22}^{-1}Q_{21} \tag{34}$$

Defining

$$v = w_2 + Q_{22}^{-1}Q_{21}w_1 \tag{35}$$

Substituting equation (34) and (35) in (33), modified performance measure is obtained as in (36)

$$J = \frac{1}{2}\int_{t_s}^{\infty} w_1^T\hat{Q}w_1 + v^TQ_{22}v \tag{36}$$

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on November 04,2026 at 22:37:49 UTC from IEEE Xplore. Restrictions apply.

Now, this performance index is to be minimized subject to constraint equation given below.

$$\dot{w_1}(t) = \hat{A}_{11} w_1(t) + A_{12} w_2(t) \tag{37}$$

Eliminating the contribution using the equation (35) ,the modified constraint equation becomes

$$\dot{w_1}(t) = A_{11} w_1(t) + A_{12} v \tag{38}$$

where

$$\hat{A} = A_{11} - A_{12} Q_{22}^{-} 1 Q_{21} \tag{39}$$

Substituting all the available values in (39),$\hat{A}$ can be obtained as

$$\hat{A} = \begin{Bmatrix} 0 & 1 \\ 0 & 0 \end{Bmatrix} \tag{40}$$

A unique positive definite solution P is guaranteed for the algebraic matrix Ricatti equation which is associated with the problem defined by equation (36)and (38) as:

$$P\hat{A} + \hat{A}^T P - P A_{12} Q_{22}^{-1} A_{12}^T P + \hat{Q} = 0 \tag{41}$$

The expression for w_2 partition is given by

$$w_2 = -Q_{22}^{-1}(A_{12}^T P + Q_{21}) w_1 \tag{42}$$

Comparing the equations (25) and (42)

$$M_1 = -Q_{22}^{-1}(A_{12}^T P + Q_{21}) \tag{43}$$

Substituting all the available values in Ricatti equation a unique positive definite solution P can be obtained as

$$P = \begin{Bmatrix} 1 & \sqrt{3} \\ \sqrt{3} & 1 \end{Bmatrix} \tag{44}$$

Inserting the value of P in equation (43)

$$M_1 = \begin{Bmatrix} 1 & \sqrt{3} \end{Bmatrix} \tag{45}$$

The switching surface S_1 will be determined fromM_1 by assuming $l_2 = I_m$. where I_m is the identity matrix.

$$S_1 T_r^T = \begin{Bmatrix} M_1 & I_m \end{Bmatrix} \tag{46}$$

$$S_1 = \begin{Bmatrix} M_1 & I_m \end{Bmatrix} * T_r \tag{47}$$

The value of is calculated as 1 using QR decomposition. Now the switching surface of the given system (16) is defined in (48) as:

$$S_1 = \begin{Bmatrix} 1 & \sqrt{3} & 1 \end{Bmatrix} \begin{Bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{Bmatrix} \tag{48}$$

Case2:
The switching surface of yaw can be calculated in the same way. The transfer function model of yaw is obtained from Equation(15) as

$$\phi = \frac{3.6}{s^3 + 6s^2 + 5s} u_2 \tag{49}$$

The phase variable form of the above equation can be written as:

$$\begin{aligned} \dot{\phi_1} &= \phi_2 \\ \dot{\phi_2} &= \phi_3 \\ \dot{\phi_3} &= -5\phi_2 - 6\phi_3 + 3.6u_2 \end{aligned} \tag{50}$$

Continuing in the same way as that of pitch angle surface design, the switching surface for yaw angle is obtained as:

$$S_2 = \begin{Bmatrix} 1 & \sqrt{3} & 1 \end{Bmatrix} \begin{Bmatrix} \phi_1 \\ \phi_2 \\ \phi_3 \end{Bmatrix} \tag{51}$$

IV. CONTROLLER DESIGN

The controller is defined using state feedback linearization [7]. In state feedback linearization, controller is designed such that the output tracks the reference signal. The reference signal is taken as in [2].

$$r_1(t) = 0.2\sin(0.1t) + 0.1\sin(0.06t) + 0.2\sin(0.04t) + 0.4 \tag{52}$$

The difference between output and reference is taken as error signal as given below

$$\begin{aligned} e_1 &= \psi_1 - r_1 \\ e_2 &= \psi_2 - \dot{r_1} \\ e_3 &= \psi_3 - \ddot{r_1} \end{aligned} \tag{53}$$

The above equation (53) when substituted in (17) yields:

$$\begin{aligned} \dot{e_1} &= e_2 \\ \dot{e_2} &= e_3 \\ \dot{e_3} &= -4.278\psi_1 - 4.78\psi_2 - .9973\psi_3 + 1.359u_1 - \dddot{r_1} \end{aligned} \tag{54}$$

Then the state feedback pitch control u_p is given by

$$\begin{aligned} u_p = \frac{1}{1.359}(4.278\psi_1 + 4.786\psi_2 \\ +0.9973\psi_3 + \dddot{r_1} - k_1 e_1 - k_2 e_2) + K_1 sign(S_1) \end{aligned} \tag{55}$$

where$K_1 = 3.5$. Similarly the yaw control is designed using state feedback linearization. The reference signal for yaw control is taken as

$$r_2(t) = 0.8\sin(0.1t) + 0.3\sin(0.05t) + 0.3\sin(0.01t) \tag{56}$$

The same procedure is repeated for yaw angle control. The state feedback yaw control u_y is given by

$$u_y = \frac{1}{3.6}(5\phi_2 + 6\phi_3 + \dddot{r_2} - k_3 e_4 - k_4 e_4) + K_2 sign(S_2) \tag{57}$$

where $K_2 = 5$

V. SIMULATION RESULTS AND DISCUSSION

The initial conditions for pitch and yaw angle are taken as 0.1 and 0 radians respectively. Fig. 3 shows the pitch angle response when the time varying sinusoidal reference signal is given. It is clear that the pitch angle is also sinusoidal and it tracks the reference signal, the maximum value of reference signal is 0.45 in radians and minimum value is -0.25(rad). The

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on November 04,2026 at 22:37:49 UTC from IEEE Xplore. Restrictions apply.

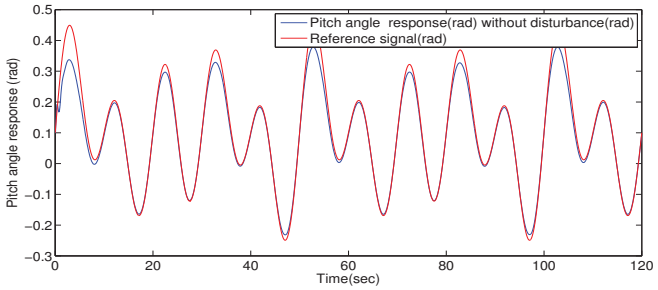


Fig. 3. pitch angle response without disturbance.

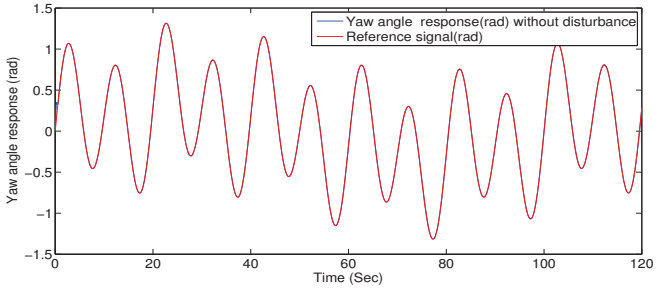


Fig. 4. yaw angle response without disturbance.

maximum and minimum values of pitch angle response are 0.38 rad and -0.232 rad. It is also clear that there is no initial over shoots and under shoots. Fig. 4 shows the yaw angle response when the time varying sinusoidal reference signal is given. It is clear that the yaw angle is also sinusoidal and it tracks the reference signal. The maximum value of reference is 1.3156 radian and minimum value is -1.3156 rad. The maximum and minimum values of yaw angle responses are 1.3144 rad and -1.3152 rad. It is also clear that there is no initial over shoots and under shoots.

Fig. 5 shows the error between the pitch angle response and reference signal. The maximum error is found to be 0.113 rad at time 3.5 sec. Fig. 6 shows error between the yaw angle response and reference signal. The maximum error is found to be 0.015 rad at time 100 sec.Fig. 7 and 8 show the pitch angle yaw angle control signals respectively.It is found that both the control values are within -2.5 and +2.5 which is needed for real time simulation.

A disturbance say $H_1 = 0.5\sin(t)+0.4$, is given to pitch and $H_2 = 5\sin(t)+0.4$ is given to yaw angle through input channel (matched disturbance). From Fig. 9 and 10, it is noted that the response is not affected much by the disturbance given.This indirectly affirms the robustness property of the sliding surface designed.

VI. Conclusion

The results of the simulation illustrate the ability of proposed design approaches, quadratic minimization technique for surface design and state feedback linearization in the controller ,

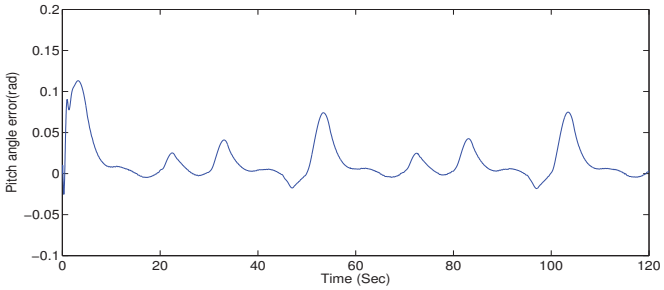


Fig. 5. pitch angle error.

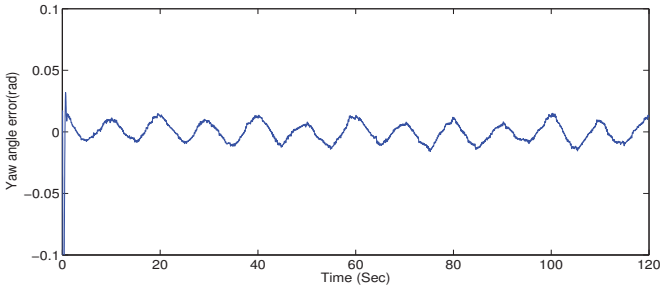


Fig. 6. yaw angle error.

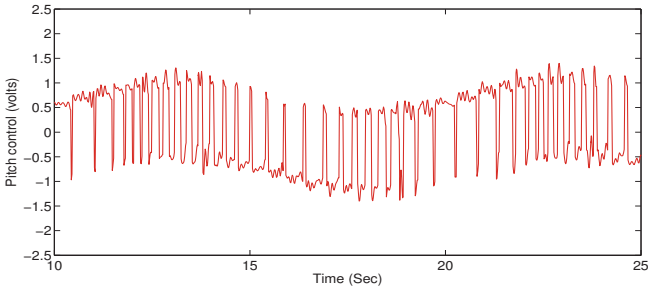


Fig. 7. pitch angle control signal.

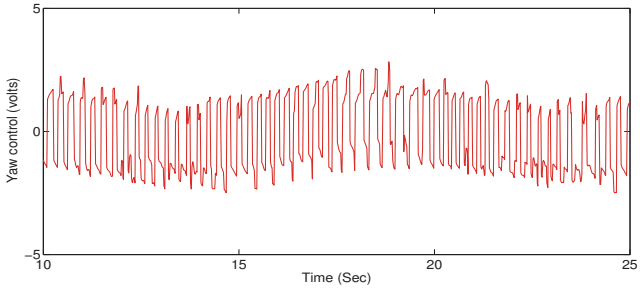


Fig. 8. Yaw control signal

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on November 04,2026 at 22:37:49 UTC from IEEE Xplore. Restrictions apply.

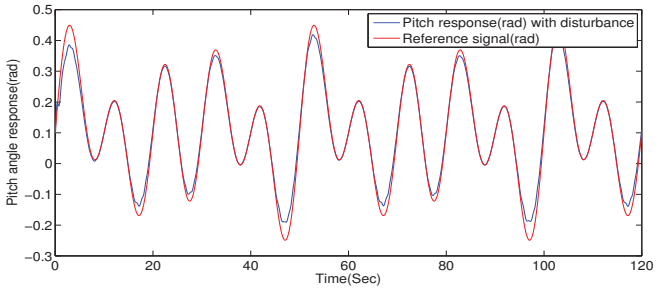


Fig. 9. pitch angle response with disturbance of $H = .5\sin(t) + 0.4$.

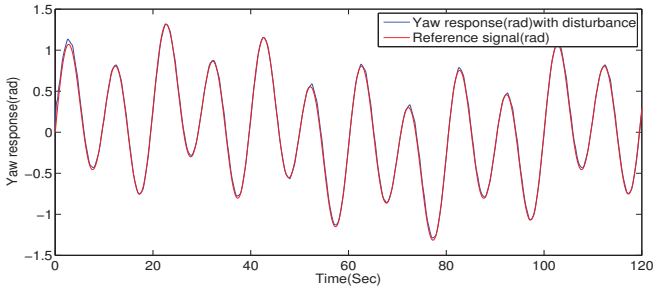


Fig. 10. yaw angle response with disturbance of $H = 5\sin(t) + 0.4$.

to track the pitch and yaw angle when time varying sinusoidal input is given. It is found that both are feasible for 2- DOF TRMS. The robustness of the surface is considerably improved by the use of quadratic minimization technique. The initial transients are considerably reduced by the use of feedback linearization.

REFERENCES

[1] TRMS 33-949S User Manual, Feedback Instruments Ltd., East Sussex,U.K., 2006.

[2] Bhanu Pratap, Abhishek Agrawal, and Shubhi Purwar " Optimal control of Twin rotor MIMO system Using Output Feedback," Proceedings of IEEE 2nd International Conference on Power Control and Embedded Systems, 2012.

[3] Lidiya John and Mija S.J., "Robust H infinity control algorithm for Twin Rotor MIMO system," Proceedings of the IEEE international conference on Advanced communication control and computing Technologies., Ramanathapuram, 2014.

[4] Jatin Kumar Pradhan , Arun Ghosh,"Design and implementation of decoupled compensation for a twin rotor multiple-input and multiple-output system," IET Control Theory Appl., Vol.7, Iss.2, pp.282-289, 2013.

[5] V.I.Utkin, Guldner J., and Shi J.X., Sliding mode control in Electro Mechanical systems, London,U.K., Taylor Francis Ltd., 2008.

[6] Chritopher Edward , Sarah K.Spergeon, Sliding Mode Control: Theory and Applications, London, U.K., Taylor Francis Ltd., 1998.

[7] Hassan K.Khalil, Nonlinear systems,Upper Saddle River,New Jersey, 3rd ed.;Prenticehall,2002.

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on November 04,2026 at 22:37:49 UTC from IEEE Xplore. Restrictions apply.